

Statistical Significance Analysis

Table S1. Statistical significance results on CSI500 RankIC

Comparison	CNN-GP (Mean ± Std)	Baseline (Mean ± Std)	Mean Diff.	Welch p	Holm-adjusted p	Mann–Whitney p	MW Holm p	Significant
CNN-GP vs AlphaForge	0.05302 ± 0.002953	0.04330 ± 0.003305	0.00972	1.880811E-06	3.761623E-06	0.000246	0.000913	Yes
CNN-GP vs D-Va	0.05302 ± 0.002953	0.03410 ± 0.003155	0.01892	5.199832E-11	1.559950E-10	0.000183	0.000913	Yes
CNN-GP vs QFR	0.05302 ± 0.002953	0.03800 ± 0.006678	0.01502	2.502733E-05	2.502733E-05	0.000246	0.000913	Yes
CNN-GP vs RiskMiner	0.05302 ± 0.002953	0.02140 ± 0.004364	0.03162	2.651493E-12	1.060597E-11	0.000183	0.000913	Yes
CNN-GP vs SFAG	0.05302 ± 0.002953	0.01880 ± 0.004195	0.03422	3.459040E-13	1.729520E-12	0.000183	0.000913	Yes

Table S2. Statistical significance results on CSI500 ALER

Comparison	CNN-GP (Mean ± Std)	Baseline (Mean ± Std)	Mean Diff.	Welch p	Holm-adjusted p	Mann–Whitney p	MW Holm p	Significant
CNN-GP vs AlphaForge	0.16480 ± 0.005587	0.08920 ± 0.002858	0.07560	4.634115E-15	9.268229E-15	0.000182	0.000898	Yes
CNN-GP vs D-Va	0.16480 ± 0.005587	0.06220 ± 0.003055	0.10260	3.046940E-17	1.218776E-16	0.000182	0.000898	Yes
CNN-GP vs QFR	0.16480 ± 0.005587	0.03801 ± 0.005625	0.12679	7.416355E-21	3.708178E-20	0.000182	0.000898	Yes
CNN-GP vs RiskMiner	0.16480 ± 0.005587	0.07120 ± 0.003078	0.09360	9.797573E-17	2.939272E-16	0.000182	0.000898	Yes
CNN-GP vs SFAG	0.16480 ± 0.005587	0.01100 ± 0.000606	0.15380	1.005444E-14	1.005444E-14	0.000180	0.000898	Yes

Table S3. Statistical significance results on CSI1000 RankIC

Comparison	CNN-GP (Mean ± Std)	Baseline (Mean ± Std)	Mean Diff.	Welch p	Holm-adjusted p	Mann–Whitney p	MW Holm p	Significant
CNN-GP vs AlphaForge	0.06460 ± 0.001678	0.05230 ± 0.002877	0.01230	9.159261E-09	9.159261E-09	0.000183	0.000913	Yes
CNN-GP vs D-Va	0.06460 ± 0.001678	0.04350 ± 0.003337	0.02110	1.135534E-10	2.271067E-10	0.000183	0.000913	Yes
CNN-GP vs QFR	0.06460 ± 0.001678	0.04430 ± 0.002677	0.02030	2.158825E-12	6.476474E-12	0.000183	0.000913	Yes
CNN-GP vs RiskMiner	0.06460 ± 0.001678	0.03370 ± 0.003349	0.03090	8.879733E-13	3.551893E-12	0.000183	0.000913	Yes
CNN-GP vs SFAG	0.06460 ± 0.001678	0.03460 ± 0.001478	0.03000	2.866179E-19	1.433090E-18	0.000183	0.000913	Yes

Table S4. Statistical significance results on CSI1000 ALER

Comparison	CNN-GP (Mean ± Std)	Baseline (Mean ± Std)	Mean Diff.	Welch p	Holm-adjusted p	Mann–Whitney p	MW Holm p	Significant
CNN-GP vs AlphaForge	0.15781 ± 0.002954	0.06320 ± 0.002378	0.09461	1.778964E-23	5.029983E-23	0.000182	0.000908	Yes
CNN-GP vs D-Va	0.15781 ± 0.002954	0.06462 ± 0.002726	0.09319	1.257496E-23	5.029983E-23	0.000182	0.000908	Yes
CNN-GP vs QFR	0.15781 ± 0.002954	0.06180 ± 0.003036	0.09601	1.480355E-23	5.029983E-23	0.000182	0.000908	Yes
CNN-GP vs RiskMiner	0.15781 ± 0.002954	0.05300 ± 0.002997	0.10481	2.661106E-24	1.330553E-23	0.000182	0.000908	Yes
CNN-GP vs SFAG	0.15781 ± 0.002954	0.08260 ± 0.002980	0.07521	9.639455E-22	9.639455E-22	0.000182	0.000908	Yes

Note: Results are reported as mean ± standard deviation over 10 independent runs. Statistical significance is assessed using a two-sided Welch’s t-test, with Holm correction applied for multiple comparisons. The Mann–Whitney U test is additionally reported as a non-parametric robustness check. A result is considered statistically significant when the Holm-adjusted Welch’s $p \leq 0.05$.

Table S5. Statistical significance results of ablation study on CSI500 RankIC

Comparison	CNN-GP (Mean ± Std)	Ablation (Mean ± Std)	Mean Diff.	Welch p	Holm-adjusted p	Mann–Whitney p	MW Holm p	Significant
CNN-GP vs CNN-XGBoost	0.05302 ± 0.002953	0.04563 ± 0.002443	0.00739	1.070822E-05	2.141645E-05	0.000440	0.000879	Yes
CNN-GP vs CNN	0.05302 ± 0.002953	0.04440 ± 0.006196	0.00862	1.620624E-03	1.620624E-03	0.009108	0.009108	Yes
CNN-GP vs GP	0.05302 ± 0.002953	0.04280 ± 0.002680	0.01022	2.171629E-07	6.514888E-07	0.000183	0.000731	Yes
CNN-GP vs XGBoost	0.05302 ± 0.002953	0.03281 ± 0.003199	0.02021	2.032530E-11	8.130121E-11	0.000183	0.000731	Yes

Table S6. Statistical significance results of ablation study on CSI500 ALER

Comparison	CNN-GP (Mean ± Std)	Ablation (Mean ± Std)	Mean Diff.	Welch p	Holm-adjusted p	Mann–Whitney p	MW Holm p	Significant
CNN-GP vs CNN-XGBoost	0.16480 ± 0.005587	0.11410 ± 0.002937	0.05070	7.204923E-13	1.440985E-12	0.000182	0.000727	Yes
CNN-GP vs CNN	0.16480 ± 0.005587	0.10530 ± 0.002985	0.05950	7.164995E-14	2.149499E-13	0.000182	0.000727	Yes
CNN-GP vs GP	0.16480 ± 0.005587	0.10810 ± 0.002072	0.05670	3.040833E-12	3.040833E-12	0.000182	0.000727	Yes
CNN-GP vs XGBoost	0.16480 ± 0.005587	0.01933 ± 0.002536	0.14547	4.948460E-18	1.979384E-17	0.000182	0.000727	Yes

Table S7. Statistical significance results of ablation study on CSI1000 RankIC

Comparison	CNN-GP (Mean ± Std)	Ablation (Mean ± Std)	Mean Diff.	Welch p	Holm-adjusted p	Mann–Whitney p	MW Holm p	Significant
CNN-GP vs CNN-XGBoost	0.06460 ± 0.001678	0.05920 ± 0.002851	0.00540	1.270775E-04	1.270775E-04	0.000183	0.000731	Yes
CNN-GP vs CNN	0.06460 ± 0.001678	0.05840 ± 0.001534	0.00620	8.813633E-08	1.762727E-07	0.000183	0.000731	Yes
CNN-GP vs GP	0.06460 ± 0.001678	0.05120 ± 0.003222	0.01340	1.904463E-08	5.713390E-08	0.000183	0.000731	Yes
CNN-GP vs XGBoost	0.06460 ± 0.001678	0.04910 ± 0.002928	0.01550	5.669691E-10	2.267876E-09	0.000183	0.000731	Yes

Table S8. Statistical significance results of ablation study on CSI1000 ALER

Comparison	CNN-GP (Mean ± Std)	Ablation (Mean ± Std)	Mean Diff.	Welch p	Holm-adjusted p	Mann–Whitney p	MW Holm p	Significant
CNN-GP vs CNN-XGBoost	0.15781 ± 0.002954	0.09848 ± 0.002795	0.05933	4.249006E-20	8.498012E-20	0.000182	0.000718	Yes
CNN-GP vs CNN	0.15781 ± 0.002954	0.12900 ± 0.000459	0.02881	9.589029E-11	9.589029E-11	0.000180	0.000718	Yes
CNN-GP vs GP	0.15781 ± 0.002954	0.07720 ± 0.002848	0.08061	1.965319E-22	5.895956E-22	0.000182	0.000718	Yes
CNN-GP vs XGBoost	0.15781 ± 0.002954	0.05404 ± 0.002422	0.10377	3.032403E-24	1.212961E-23	0.000182	0.000718	Yes